

Expectation Maximization (EM): A Rigorous and Intuitive Treatment

The Expectation Maximization (EM) algorithm provides a principled way to perform maximum likelihood estimation when the observed data is generated through a process involving hidden (latent) variables. The central difficulty in such problems is that we do not have access to all variables involved in the data generation mechanism.

Suppose we observe data

$$X = \{x_1, x_2, \dots, x_n\},$$

but assume that each observation is associated with a hidden variable

$$Z = \{z_1, z_2, \dots, z_n\}.$$

We posit a parametric model

$$p(X, Z \mid \theta),$$

and aim to estimate the parameters by maximizing the likelihood

$$\hat{\theta} = \arg \max_{\theta} \log p(X \mid \theta).$$

However, since

$$p(X \mid \theta) = \sum_Z p(X, Z \mid \theta),$$

we obtain

$$\log p(X \mid \theta) = \log \sum_Z p(X, Z \mid \theta),$$

and this is precisely where the difficulty lies: the logarithm of a sum does not simplify, making direct optimization analytically intractable.

A key idea is to introduce an auxiliary distribution $q(Z)$ satisfying $\sum_Z q(Z) = 1$. This allows us to rewrite the log-likelihood as

$$\log p(X \mid \theta) = \log \sum_Z q(Z) \frac{p(X, Z \mid \theta)}{q(Z)}.$$

This manipulation does not change the expression, but it enables a crucial transformation. Since the logarithm is concave, Jensen's inequality gives

$$\log p(X \mid \theta) \geq \sum_Z q(Z) \log \frac{p(X, Z \mid \theta)}{q(Z)}.$$

We denote the right-hand side by

$$\mathcal{L}(q, \theta) = \sum_Z q(Z) \log p(X, Z \mid \theta) - \sum_Z q(Z) \log q(Z),$$

which serves as a lower bound to the true log-likelihood:

$$\log p(X \mid \theta) \geq \mathcal{L}(q, \theta).$$

This expression has two important components. The first term,

$$\sum_Z q(Z) \log p(X, Z \mid \theta),$$

can be interpreted as the expected complete-data log-likelihood: if the hidden variables were known, we would directly maximize $\log p(X, Z \mid \theta)$, and here we replace the unknown Z by an expectation under $q(Z)$. The second term,

$$- \sum_Z q(Z) \log q(Z),$$

measures the uncertainty (entropy) of the distribution $q(Z)$.

An important observation is that this entropy term depends only on $q(Z)$ and not on θ . Consequently, when optimizing with respect to θ , this term becomes a constant and plays no role in parameter updates. From an engineering viewpoint, it simply reflects how uncertain our current belief about the hidden variables is; since we do not change this belief during parameter optimization, it does not affect that step.

A natural question at this point is: *when does the inequality become an equality?* Understanding this is crucial, because it explains the logic behind the E-step.

Recall that we applied Jensen's inequality in the form

$$\log \sum_Z q(Z) \frac{p(X, Z \mid \theta)}{q(Z)} \geq \sum_Z q(Z) \log \frac{p(X, Z \mid \theta)}{q(Z)}.$$

At this point, it is important to understand precisely when Jensen's inequality becomes an equality.

Recall the general form of Jensen's inequality for a concave function ϕ :

$$\phi(\mathbb{E}[Y]) \geq \mathbb{E}[\phi(Y)].$$

In our case, $\phi = \log$ and

$$Y(Z) = \frac{p(X, Z \mid \theta)}{q(Z)},$$

so we are comparing

$$\log \mathbb{E}_q[Y(Z)] \quad \text{and} \quad \mathbb{E}_q[\log Y(Z)].$$

Key fact. For a strictly concave function such as $\log$, Jensen's inequality becomes an equality *if and only if* the random variable inside the expectation is constant (with probability one under q). That is,

$$Y(Z) = \text{constant for all } Z \text{ such that } q(Z) > 0.$$

Why is this true? The logarithm is strictly concave, meaning its graph lies strictly below any straight line connecting two distinct points. As a result:

- If $Y(Z)$ takes different values for different Z , then averaging before applying $\log$ is strictly better than averaging after applying $\log$.

- Equality can only occur if there is no variability at all, i.e., $Y(Z)$ takes the same value everywhere.

Intuition. Jensen’s inequality compares:

- “Take an average first, then apply log”
- “Apply log first, then take an average”

For a concave function like log, averaging smooths out variability and increases the value. The only way the two procedures can give the same result is if there is nothing to smooth out—in other words, if all values being averaged are identical.

Applying this to our case. We therefore require

$$\frac{p(X, Z \mid \theta)}{q(Z)} = C \quad \text{for all } Z \text{ with } q(Z) > 0,$$

for some constant C independent of Z .

This condition is what leads directly to

$$q(Z) \propto p(X, Z \mid \theta),$$

and hence to the posterior distribution after normalization.

Summary. The equality condition in Jensen’s inequality is not a technical detail—it is the central reason why the optimal choice of $q(Z)$ in EM is the posterior distribution.

In our case, this means that the quantity

$$\frac{p(X, Z \mid \theta)}{q(Z)}$$

must be the same for all values of Z for which $q(Z) > 0$. In other words, there must exist a constant C (independent of Z) such that

$$\frac{p(X, Z \mid \theta)}{q(Z)} = C.$$

Rearranging this expression gives

$$q(Z) = \frac{1}{C} p(X, Z \mid \theta).$$

Now we determine the constant C using the fact that $q(Z)$ is a probability distribution:

$$\sum_Z q(Z) = 1.$$

Substituting the expression above,

$$\sum_Z \frac{1}{C} p(X, Z \mid \theta) = 1 \quad \Rightarrow \quad \frac{1}{C} \sum_Z p(X, Z \mid \theta) = 1.$$

But we recognize that

$$\sum_Z p(X, Z \mid \theta) = p(X \mid \theta),$$

so this gives

$$\frac{1}{C} p(X | \theta) = 1 \Rightarrow C = p(X | \theta).$$

Substituting back, we obtain

$$q(Z) = \frac{p(X, Z | \theta)}{p(X | \theta)}.$$

By definition, this is precisely the posterior distribution of the latent variables:

$$q(Z) = p(Z | X, \theta).$$

Interpretation. The lower bound becomes exact when our chosen distribution $q(Z)$ matches the true posterior distribution of the hidden variables given the observed data and current parameters. In that case, the “gap” introduced by Jensen’s inequality disappears entirely.

Key insight. This result explains the E-step of the EM algorithm: the best possible choice of $q(Z)$, in the sense of making the bound as tight as possible, is to set it equal to the posterior $p(Z | X, \theta)$.

At this point, the optimization problem becomes manageable by alternating between improving q and improving θ . To formalize the parameter update, it is useful to define the function

$$Q(\theta | \theta^{(t)}) = \mathbb{E}_{Z \sim p(Z | X, \theta^{(t)})} [\log p(X, Z | \theta)].$$

This function deserves careful interpretation: the expectation is taken with respect to the current parameter $\theta^{(t)}$, while the parameter inside the logarithm is the new variable θ that we wish to optimize. In other words, we assume the hidden variables behave according to the current model, and then choose parameters that best explain this inferred structure.

The EM algorithm now follows a simple iterative pattern. Starting from an initial guess, we alternate between:

E-step:

$$q^{(t+1)}(Z) = p(Z | X, \theta^{(t)}),$$

which computes the posterior distribution of the hidden variables, often interpreted as soft assignments or responsibilities.

M-step:

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta | \theta^{(t)}),$$

which updates the parameters by maximizing the expected complete-data log-likelihood.

These two steps correspond, respectively, to inference and learning, and together form a self-consistent refinement loop.

It is also important to understand why this procedure converges. After the E-step, the lower bound becomes tight at the current parameter value:

$$\mathcal{L}(q^{(t+1)}, \theta^{(t)}) = \log p(X | \theta^{(t)}).$$

The M-step then increases this bound:

$$\mathcal{L}(q^{(t+1)}, \theta^{(t+1)}) \geq \mathcal{L}(q^{(t+1)}, \theta^{(t)}).$$

Since the true log-likelihood always lies above the bound,

$$\log p(X \mid \theta^{(t+1)}) \geq \mathcal{L}(q^{(t+1)}, \theta^{(t+1)}),$$

and combining these relations yields

$$\log p(X \mid \theta^{(t+1)}) \geq \log p(X \mid \theta^{(t)}).$$

Thus, the likelihood is guaranteed to be non-decreasing, and because it is bounded above, the sequence converges (typically to a local optimum).

Mixture of Two Binomials: Fully Explained Example

We now revisit the classical two-coin problem, but this time we develop it carefully from the probabilistic model all the way to the numerical updates.

Suppose we have two coins:

$$\theta_A = P(\text{Head} \mid A), \quad \theta_B = P(\text{Head} \mid B).$$

An experiment is performed as follows:

- One of the two coins is selected (unknown to us)
- The selected coin is tossed 10 times
- Only the number of heads is recorded

Repeating this experiment gives the observed data:

$$X = \{8, 7, 9, 2, 3, 1\}.$$

Step 1: Probabilistic Model

If a coin with bias θ is tossed 10 times, the probability of observing x heads is given by the Binomial distribution:

$$\text{Bin}(x \mid 10, \theta) = \binom{10}{x} \theta^x (1 - \theta)^{10-x}.$$

Thus, each observation x_i is generated from a mixture:

$$p(x_i) = \pi \text{Bin}(x_i \mid 10, \theta_A) + (1 - \pi) \text{Bin}(x_i \mid 10, \theta_B),$$

where π is the probability of choosing coin A .

Hidden variable. Introduce

$$z_i = \begin{cases} 1 & \text{if coin A was used} \\ 0 & \text{if coin B was used} \end{cases}$$

We do not observe z_i , and this is what makes the problem non-trivial.

Initial guess. We start with:

$$\theta_A^{(0)} = 0.6, \quad \theta_B^{(0)} = 0.4, \quad \pi^{(0)} = 0.5.$$

Iteration 1

Deriving the E-step (Responsibilities) Carefully

We want to compute

$$\gamma_i = P(z_i = 1 \mid x_i),$$

that is, the probability that the i -th observation was generated by coin A .

We apply Bayes' rule:

$$P(z_i = 1 \mid x_i) = \frac{P(x_i \mid z_i = 1) P(z_i = 1)}{P(x_i)}.$$

Now identify each term:

- $P(z_i = 1) = \pi$ (probability of choosing coin A)
- $P(x_i \mid z_i = 1) = \text{Bin}(x_i \mid 10, \theta_A)$
- $P(x_i)$ is obtained using total probability:

$$P(x_i) = \pi \text{Bin}(x_i \mid 10, \theta_A) + (1 - \pi) \text{Bin}(x_i \mid 10, \theta_B)$$

Substituting, we obtain:

$$\gamma_i = \frac{\pi \cdot \text{Bin}(x_i \mid 10, \theta_A)}{\pi \cdot \text{Bin}(x_i \mid 10, \theta_A) + (1 - \pi) \cdot \text{Bin}(x_i \mid 10, \theta_B)}.$$

Interpretation. This is a normalized likelihood:

- Numerator: how strongly coin A explains x_i
- Denominator: total explanation from both coins

Thus, γ_i is a *soft assignment* of x_i to coin A .

Deriving the M-step Updates via Differentiation

We now maximize the expected complete-data log-likelihood:

$$Q(\theta) = \sum_{i=1}^n \left[\gamma_i \log \pi + (1 - \gamma_i) \log(1 - \pi) + \gamma_i \log \text{Bin}(x_i \mid 10, \theta_A) + (1 - \gamma_i) \log \text{Bin}(x_i \mid 10, \theta_B) \right].$$

We optimize each parameter separately.

1. Update for π

We focus on terms involving π :

$$Q(\pi) = \sum_i [\gamma_i \log \pi + (1 - \gamma_i) \log(1 - \pi)].$$

Differentiate:

$$\frac{dQ}{d\pi} = \sum_i \left(\frac{\gamma_i}{\pi} - \frac{1 - \gamma_i}{1 - \pi} \right).$$

Set to zero:

$$\sum_i \frac{\gamma_i}{\pi} = \sum_i \frac{1 - \gamma_i}{1 - \pi}.$$

Multiply both sides:

$$(1 - \pi) \sum_i \gamma_i = \pi \sum_i (1 - \gamma_i).$$

Simplify:

$$\sum_i \gamma_i = n\pi.$$

Thus:

$$\boxed{\pi = \frac{1}{n} \sum_i \gamma_i}$$

Interpretation: π is simply the fraction of data assigned (softly) to coin A.

2. Update for θ_A

Recall:

$$\text{Bin}(x \mid 10, \theta_A) = \binom{10}{x} \theta_A^x (1 - \theta_A)^{10-x}.$$

Taking logarithm:

$$\log \text{Bin}(x \mid 10, \theta_A) = \log \binom{10}{x} + x \log \theta_A + (10 - x) \log(1 - \theta_A).$$

Ignoring constants, we maximize:

$$Q(\theta_A) = \sum_i \gamma_i \left[x_i \log \theta_A + (10 - x_i) \log(1 - \theta_A) \right].$$

Differentiate:

$$\frac{dQ}{d\theta_A} = \sum_i \gamma_i \left(\frac{x_i}{\theta_A} - \frac{10 - x_i}{1 - \theta_A} \right).$$

Set to zero:

$$\sum_i \gamma_i \frac{x_i}{\theta_A} = \sum_i \gamma_i \frac{10 - x_i}{1 - \theta_A}.$$

Multiply:

$$(1 - \theta_A) \sum_i \gamma_i x_i = \theta_A \sum_i \gamma_i (10 - x_i).$$

Expand right side:

$$\theta_A \left(10 \sum_i \gamma_i - \sum_i \gamma_i x_i \right).$$

Rearranging gives:

$$\sum_i \gamma_i x_i = 10\theta_A \sum_i \gamma_i.$$

Thus:

$$\boxed{\theta_A = \frac{\sum_i \gamma_i x_i}{\sum_i \gamma_i \cdot 10}}$$

Interpretation:

- Numerator = expected number of heads from coin A
- Denominator = expected number of tosses from coin A

So θ_A is a weighted empirical probability of heads.

3. Update for θ_B

By symmetry:

$$\theta_B = \frac{\sum_i (1 - \gamma_i) x_i}{\sum_i (1 - \gamma_i) \cdot 10}$$

Interpretation: Same logic, but using responsibility of coin B.

Final Insight

Each update has a very natural meaning:

- γ_i = probability that coin A generated x_i
- π = average responsibility of coin A
- θ_A = weighted fraction of heads for coin A
- θ_B = weighted fraction of heads for coin B

Thus EM converts a hidden-variable problem into a sequence of

“compute probabilities” $\rightarrow$ “compute weighted averages”

steps.

Numerical Computation (Putting Everything Together)

We now apply the derived formulas step by step using the dataset:

$$X = \{8, 7, 9, 2, 3, 1\}$$

with initial parameters:

$$\theta_A^{(0)} = 0.6, \quad \theta_B^{(0)} = 0.4, \quad \pi^{(0)} = 0.5.$$

Iteration 1

E-step: Computing γ_i

Recall:

$$\gamma_i = \frac{\pi \cdot \text{Bin}(x_i \mid 10, \theta_A)}{\pi \cdot \text{Bin}(x_i \mid 10, \theta_A) + (1 - \pi) \cdot \text{Bin}(x_i \mid 10, \theta_B)}.$$

Since $\binom{10}{x_i}$ cancels, we compute:

$$\theta^x (1 - \theta)^{10-x}.$$

Example: $x = 8$

$$L_A = 0.6^8 \cdot 0.4^2 \approx 0.0060, \quad L_B = 0.4^8 \cdot 0.6^2 \approx 0.0007.$$

$$\gamma_1 = \frac{0.5 \times 0.0060}{0.5 \times 0.0060 + 0.5 \times 0.0007} \approx 0.92.$$

Repeating for all observations:

x_i	γ_i
8	0.92
7	0.85
9	0.97
2	0.10
3	0.20
1	0.05

Check: total responsibility

$$\sum_i \gamma_i = 0.92 + 0.85 + 0.97 + 0.10 + 0.20 + 0.05 = 3.09.$$

M-step: Updating parameters

Update for π

$$\pi^{(1)} = \frac{1}{6} \sum_i \gamma_i = \frac{3.09}{6} \approx 0.515.$$

Update for θ_A

Compute weighted heads:

$$\sum \gamma_i x_i = 0.92 \cdot 8 + 0.85 \cdot 7 + 0.97 \cdot 9 + 0.10 \cdot 2 + 0.20 \cdot 3 + 0.05 \cdot 1.$$

$$= 7.36 + 5.95 + 8.73 + 0.20 + 0.60 + 0.05 = 22.89.$$

Weighted tosses:

$$\sum \gamma_i \cdot 10 = 3.09 \cdot 10 = 30.9.$$

$$\theta_A^{(1)} = \frac{22.89}{30.9} \approx 0.74.$$

Update for θ_B

Total heads:

$$\sum x_i = 30.$$

Heads for B:

$$30 - 22.89 = 7.11.$$

Tosses for B:

$$60 - 30.9 = 29.1.$$

$$\theta_B^{(1)} = \frac{7.11}{29.1} \approx 0.24.$$

End of Iteration 1

$$\theta_A = 0.74, \quad \theta_B = 0.24, \quad \pi = 0.515.$$

Interpretation

- Coin A becomes more strongly biased toward heads
- Coin B becomes more strongly biased toward tails
- Responsibilities will now become more extreme

Iteration 2

E-step (updated parameters)

Now:

$$\theta_A = 0.74, \quad \theta_B = 0.24, \quad \pi = 0.515.$$

Example: $x = 8$

$$L_A = 0.74^8 \cdot 0.26^2 \approx 0.012, \quad L_B = 0.24^8 \cdot 0.76^2 \approx \text{very small.}$$

$$\gamma_1 \approx 0.97.$$

Updated responsibilities:

x_i	$\gamma_i^{(2)}$
8	0.97
7	0.93
9	0.99
2	0.03
3	0.07
1	0.01

M-step

$$\sum \gamma_i \approx 3.00 \quad \Rightarrow \quad \pi^{(2)} \approx 0.50.$$

$$\sum \gamma_i x_i \approx 23.8, \quad \sum \gamma_i \cdot 10 = 30.$$

$$\theta_A^{(2)} \approx 0.79, \quad \theta_B^{(2)} \approx 0.21.$$

Final Interpretation

- Responsibilities move toward 0 or 1 (hard assignments)
- Parameters stabilize after a few iterations
- High-head observations belong to coin A

- Low-head observations belong to coin B

Key Insight

Even without observing which coin was used, EM successfully recovers:

- the bias of each coin
- the fraction of times each coin is used

This is achieved by iteratively:

soft assignment (E-step) $\rightarrow$ weighted estimation (M-step).